

Does Momentum Investing Generate Excess Returns in the Indian Stock Market?

A Quantitative Research Framework for Indian Equities

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NSE / BSE • 6-month and 12-month momentum • Winner-minus-Loser framework

RESEARCH POSITION

Published index evidence supports historical momentum in India, but the final investment claim requires an independent, transaction-cost-adjusted backtest.

01. Executive Summary

Momentum investing asks whether recent winners continue to outperform recent losers. This research examines that question in the Indian equity market using a systematic, rules-based framework rather than discretionary stock selection.

The proposed study focuses on 6-month and 12-month price momentum. At each monthly formation date, eligible securities are ranked by their prior returns. The top 20% form a Winner portfolio and the bottom 20% form a Loser portfolio. The difference between the two portfolios is the Winner-minus-Loser (WML) return.

Performance is evaluated against the Nifty 50 and Nifty 500 and through several risk measures, including CAPM Alpha, factor-adjusted alpha, Sharpe Ratio, Maximum Drawdown and statistical significance. The framework is designed to distinguish a genuinely persistent momentum effect from returns that can be explained by broad market exposure or other common risk characteristics.

Published Indian index evidence is used as contextual evidence rather than as a substitute for the proposed independent backtest. In particular, the Nifty200 Momentum 30 provides a useful real-world reference because its methodology incorporates 6-month and 12-month volatility-adjusted momentum.

The central conclusion of the research framework is deliberately cautious: historical evidence can support the existence of momentum without implying guaranteed future alpha. A final investment conclusion requires an independent stock-level backtest that addresses survivorship bias, look-ahead bias, liquidity and transaction costs.

02. Research Question & Objectives

The primary research question is: Does a systematic price-momentum strategy generate statistically significant and economically meaningful excess returns in the Indian stock market?

The research is designed to answer this question using a reproducible portfolio-construction process. The emphasis is on whether momentum remains meaningful after comparison with broad market benchmarks and after controlling for common sources of risk.

2.1 Research Objectives

- 1. Measure the subsequent performance of Indian equities ranked by their previous 6-month and 12-month returns.
- 2. Construct Winner, Loser and Winner-minus-Loser portfolios using clearly defined portfolio rules.
- 3. Compare momentum performance with the Nifty 50 and Nifty 500 benchmarks.
- 4. Estimate risk-adjusted performance using CAPM and broader factor models where reliable factor data are available.

- 5. Evaluate statistical significance using confidence intervals, t-statistics and appropriate standard errors.
- 6. Examine whether the results remain stable across alternative formation periods, holding periods, liquidity filters and sub-periods.
- 7. Estimate how transaction costs, turnover and other market frictions could change gross momentum returns into realistic net returns.

2.2 Hypotheses

H0: The average Winner-minus-Loser return is equal to zero; systematic price momentum does not generate a distinct return premium.

H1: The average Winner-minus-Loser return is different from zero and, if positive and statistically significant, provides evidence consistent with a momentum premium.

A second hypothesis concerns risk adjustment: if WML remains positive after controlling for market and common factor exposures, the evidence for an independent momentum effect becomes stronger.

03. Academic Foundation

The modern momentum literature is strongly associated with Jegadeesh and Titman (1993), who documented return continuation in which securities with strong recent performance tended to continue outperforming weaker performers over intermediate horizons.

Rouwenhorst (1998) extended the evidence internationally, showing that momentum was not limited to the United States. This international perspective is important for India because emerging markets differ from developed markets in liquidity, investor composition, information flow and trading frictions.

Indian research has also reported evidence consistent with momentum. Sehgal and Jain (2015) examined price, earnings and revenue momentum; Maheshwari and Dhankar (2017) considered momentum alongside characteristics such as size and illiquidity; and Agarwalla, Jacob and Varma (2017) placed size, value and momentum within a broader Indian factor framework.

The literature also suggests that anomaly strength can vary over time. Sharma, Subramaniam and Sehgal (2021) highlight the importance of examining whether historical anomalies remain stable across periods and specifications. This motivates the robustness analysis proposed in this paper.

3.1 Why India Is an Important Test Market

India provides a useful setting for studying momentum because it combines a large and increasingly institutionalised equity market with meaningful differences in liquidity, investor participation and market access across securities.

The Indian market also provides a practical implementation challenge. A strategy can look attractive before costs while becoming less compelling once bid-ask spreads, taxes, turnover and market impact are included. Therefore, the research treats implementation frictions as part of the question rather than as an afterthought.

04. Data & Research Design

The preferred universe is a broad Indian equity set, such as the Nifty 500, combined with historical constituent information wherever it can be obtained reliably. A broad universe provides sufficient cross-sectional variation for ranking and reduces dependence on a small group of large companies.

The ideal dataset should contain stock identifier, date, adjusted or total-return-consistent price, trading volume, historical index membership and benchmark returns. Daily data can be converted

to monthly formation observations, provided corporate actions and missing observations are handled consistently.

The final dataset should explicitly address survivorship bias, look-ahead bias, delistings, ticker changes, corporate actions, liquidity and missing data. These are not technical details: each can materially alter a momentum backtest if handled incorrectly.

4.1 Data Sources

Potential primary sources include NSE historical market data and Nifty Indices publications for benchmark and index information. For a research-quality stock-level backtest, a survivorship-bias-aware academic or institutional dataset is preferable where available.

Published index factsheets and whitepapers are useful for contextual validation. They should not, however, be treated as equivalent to the independent Winner-minus-Loser strategy tested in this research.

4.2 Formation and Holding Periods

The baseline design uses monthly portfolio formation. At each formation date, the previous 6-month and 12-month stock returns are calculated using only information that was available before the portfolio was formed.

The baseline portfolio takes the strongest 20% of eligible securities as Winners and the weakest 20% as Losers. Equal weighting is recommended for the initial test because it provides a simple and transparent portfolio rule.

Holding periods should include 1, 3, 6 and 12 months. The 6-month formation period is a practical primary specification, while the 12-month formation period provides an important robustness test. A 3-month formation period can be added to determine whether the result is concentrated at shorter horizons.

05. Portfolio Construction

Step 1 — Calculate formation-period returns for every eligible security.

Step 2 — Rank securities cross-sectionally from strongest to weakest based on prior performance.

Step 3 — Select the top 20% as Winners and bottom 20% as Losers.

Step 4 — Assign equal initial weights unless an alternative weighting scheme is being tested.

Step 5 — Hold the portfolios for the specified holding period and calculate subsequent returns.

Step 6 — Calculate WML as Winner return minus Loser return.

5.1 Core Formula

$$WML_t = R_{Winner,t} - R_{Loser,t}$$

A positive WML means that the securities that previously performed well subsequently outperformed the securities that previously performed poorly. A negative WML indicates the opposite.

The formation return must never overlap with information from the future holding period. This separation is essential for preventing look-ahead bias.

5.2 Formation × Holding Matrix

The recommended research matrix is designed to test whether the momentum effect is robust across reasonable horizons rather than being dependent on one selected parameter.

06. Performance Measurement

Raw return is only the first layer of analysis. A serious factor study must determine whether performance is large relative to risk, whether it remains after controlling for common factors and whether it is statistically distinguishable from zero.

6.1 CAPM Alpha

The CAPM regression can be written as: $R_{WML} - R_f = \alpha + \beta(R_M - R_f) + \varepsilon$.

The coefficient α represents the portion of WML performance not explained by market exposure under the CAPM. A positive, statistically significant alpha would strengthen the case that momentum is not simply a leveraged market exposure.

6.2 Factor-Adjusted Performance

Where reliable Indian factor series are available, a Fama-French-style model can be used to control for market, size and value exposures. Additional factors may be considered only when their definitions and data quality are transparent.

This step matters because a momentum portfolio can unintentionally concentrate in small, value-oriented or otherwise distinctive companies. If the apparent momentum return disappears after factor adjustment, the interpretation should be more cautious.

6.3 Risk and Return Metrics

Average Return — measures the mean strategy return.

Excess Return — compares the strategy with the risk-free rate or a selected benchmark.

Sharpe Ratio — evaluates return relative to volatility.

Maximum Drawdown — measures the largest peak-to-trough decline.

CAPM / Factor Alpha — estimates return beyond common risk exposures.

t-statistic and p-value — evaluate statistical evidence against a zero-return null hypothesis.

07. Published Historical Evidence

The following evidence is included as an institutional reference point. It is not presented as the empirical result of Tanmay Gupta's independent stock-level backtest.

The Nifty200 Momentum 30 is particularly relevant because it is constructed using 6-month and 12-month volatility-adjusted price returns. Its published historical performance therefore provides context for the broader question of whether momentum has worked in Indian equities.

7.1 Published Evidence Snapshot

Nifty200 Momentum 30: 5-year TRI CAGR of 14.04% and since-inception TRI CAGR of 18.79% as of April 30, 2026, with a base date of April 1, 2005.

A Nifty Indices 2026 whitepaper reported 19.19% annualized performance for Nifty200 Momentum 30 versus 15.08% for the Nifty 200 through February 27, 2026.

These figures are useful supporting evidence, but they do not replace the proposed WML backtest. The index has its own selection, volatility adjustment, weighting and rebalancing methodology.

08. Drivers of Momentum

Behavioral underreaction is one possible explanation. Investors may incorporate new information gradually, causing prices to adjust over time rather than immediately. If this adjustment is persistent, recent winners may continue to outperform recent losers.

Limits to arbitrage provide a second explanation. Even when investors identify a pricing pattern, trading against it can involve uncertainty, liquidity constraints, short-selling restrictions and market impact. These frictions can allow a return pattern to persist.

A third interpretation is risk-based: momentum portfolios may contain systematic exposures that compensate investors for risks that are not fully captured by a simple market model. This is why factor-adjusted testing is necessary.

09. Market Frictions & Transaction Costs

Gross momentum returns are not equivalent to investor returns. A strategy that frequently changes its holdings can generate substantial turnover, and every trade can incur direct or indirect costs.

Relevant Indian implementation frictions include Securities Transaction Tax (STT), bid-ask spreads, brokerage and exchange charges, market impact and the cost of trading less-liquid securities.

The central implementation relationship is: $\text{Net Return} = \text{Gross Return} - \text{Trading Costs} - \text{Market Impact} - \text{Other Applicable Costs}$.

A robust study should therefore report both gross and transaction-cost-adjusted performance under several reasonable cost assumptions. The strategy should not be judged solely on its frictionless historical return.

10. Robustness Checks

Formation periods: compare 3-, 6- and 12-month ranking windows.

Holding periods: compare 1-, 3-, 6- and 12-month holding windows.

Portfolio breakpoints: test whether the result depends on the top/bottom 20% definition.

Liquidity filters: examine whether the premium disappears when the least-liquid securities are removed.

Sub-period analysis: compare different market regimes, including strong bull markets, corrections and stressed periods.

Transaction-cost sensitivity: evaluate whether the result remains economically meaningful after increasing cost assumptions.

Benchmark-relative analysis: compare WML with both Nifty 50 and Nifty 500 rather than relying on a single market benchmark.

10.1 Bias and Data-Quality Controls

Survivorship bias must be controlled by using historical constituents where possible. Using only today's surviving companies can make historical performance look artificially strong.

Look-ahead bias must be prevented by ensuring that portfolio formation uses only information available before the formation date.

Corporate actions, missing observations, delistings and ticker changes should be processed consistently so that calculated returns represent actual economic price changes rather than data artefacts.

11. Expected Empirical Output

The final analysis should report descriptive statistics for stock returns and portfolio returns, Winner and Loser performance, WML performance, risk-adjusted metrics and statistical tests.

Recommended figures include cumulative wealth for Winner, Loser, WML and benchmark portfolios; the distribution of WML returns; rolling momentum performance; drawdown comparisons; and a formation-period × holding-period heatmap.

No empirical regression coefficient, alpha, Sharpe Ratio or p-value should be invented. Until the actual historical dataset is processed, those numerical results should be labelled as to be calculated.

12. Conclusion

The existing academic literature and published Indian index evidence provide meaningful support for momentum as a historical feature of Indian equities. The evidence is consistent with the idea that recent winners can continue to outperform recent losers over intermediate horizons.

However, historical momentum should not be described as guaranteed alpha. Its strength can vary across time, market conditions, portfolio construction rules and risk models, while implementation costs can materially reduce the gross return available to an investor.

The strongest final research conclusion should therefore come from the independent stock-level backtest. If WML remains positive, statistically significant, economically meaningful and reasonably robust after risk adjustment and realistic transaction costs, the evidence for an investable Indian momentum premium becomes substantially stronger.

The research ultimately distinguishes between two claims: momentum has historically existed in Indian equities, and momentum can be profitably implemented today. The first can be supported by published evidence; the second requires a transparent, transaction-cost-adjusted empirical test.

13. References

- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1), 65–91.
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14. Appendix — Research Workflow

- 1. Obtain historical stock prices and constituent information.
- 2. Clean identifiers, corporate actions and missing observations.
- 3. Construct monthly observations and calculate 6- and 12-month formation returns.
- 4. Rank stocks cross-sectionally at each formation date.
- 5. Form Winner and Loser portfolios using the pre-specified breakpoint.
- 6. Calculate holding-period returns and WML returns.
- 7. Compare WML with Nifty 50 and Nifty 500.
- 8. Calculate volatility, Sharpe Ratio and Maximum Drawdown.
- 9. Estimate CAPM and factor-adjusted alpha.
- 10. Test statistical significance and confidence intervals.
- 11. Run formation/holding, liquidity and sub-period robustness checks.
- 12. Apply transaction-cost assumptions and report net results.
- 13. Document limitations and interpret the evidence without overstating causality.

05. Portfolio Construction — Research Matrix

Formation	Holding	Purpose
3 months	1 / 3 / 6 / 12 months	Shorter-horizon robustness
6 months	1 / 3 / 6 / 12 months	Primary medium-term momentum test
12 months	1 / 3 / 6 / 12 months	Longer-horizon robustness

07. Published Evidence Snapshot

Evidence	Reported result	Research interpretation
Nifty200 Momentum 30 — 5-year TRI CAGR	14.04%	Published index evidence as of 30 Apr 2026
Nifty200 Momentum 30 — since inception	18.79%	Published CAGR using 1 Apr 2005 base date
Nifty200 Momentum 30 vs Nifty 200	19.19% vs 15.08%	Annualized performance through 27 Feb 2026
Methodology	6M + 12M volatility-adjusted momentum	Institutional index evidence; not the independent WML backtest

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- 1. Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *The Journal of Finance*, 48(1), 65–91.
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- 7. NSE Indices Limited. (2026). Nifty200 Momentum 30 Factsheet — April 30, 2026.

- 8. NSE Indices Limited. (2026). The Science of Momentum: Capturing the Upside of Market Movements. Momentum Strategy Whitepaper.

14. Appendix — Research Workflow

- 1. Obtain historical stock prices and historical constituent information.
- 2. Clean identifiers, corporate actions, missing observations and delistings.
- 3. Convert the data to monthly formation observations.
- 4. Calculate 6-month and 12-month formation returns.
- 5. Rank securities cross-sectionally at each formation date.
- 6. Form Winner and Loser portfolios using the pre-specified 20% breakpoints.
- 7. Calculate holding-period returns and WML.
- 8. Compare WML with the Nifty 50 and Nifty 500.
- 9. Calculate volatility, Sharpe Ratio and Maximum Drawdown.
- 10. Estimate CAPM and factor-adjusted alpha where data permit.
- 11. Test statistical significance and confidence intervals.
- 12. Run formation/holding, liquidity, sub-period and transaction-cost robustness checks.
- 13. Report gross and net results separately.
- 14. Interpret findings conservatively and distinguish historical evidence from future investment expectations.